

```
In [1]: def greet():  
        print('hello')
```

```
In [2]: def greet():# greet-function name  
        print('hello good morning')  
        greet() # call the function
```

hello good morning

```
In [4]: def greet():  
        print('hello good morning')  
        greet()  
        def greet():  
            print('hello good morning')  
        greet()  
        def gret():  
            print('hello good morning')  
        greet()
```

hello good morning
hello good morning
hello good morning

```
In [5]: def greet():  
        print('hello good morning')  
        greet()  
        greet()  
        greet()
```

hello good morning
hello good morning
hello good morning

```
In [6]: def add(x,y):# x and y arwe formal argumnets  
        c=x+y  
        print(c)  
        add(5,6) #5 and 6 are actual arguments
```

11

```
In [8]: def add(x,y):  
        c=x+y  
        print(c)  
        r=add(5,6)  
        print(r)  
        print(type(r))
```

11
None
<class 'NoneType'>

```
In [9]: def add(x,y):  
        c=x+y  
        return c  
        r=add(5,6)  
        print(r)  
        print(type(r))
```

```
11
<class 'int'>
```

```
In [11]: def add(x,y):
          c=x+y
          return c
          add(3,4,5)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[11], line 4
      2     c=x+y
      3     return c
----> 4 add(3,4,5)

TypeError: add() takes 2 positional arguments but 3 were given
```

```
In [13]: def add(x,y,z):
          c=x+y+z
          return c
          add(3,4,5)
```

```
Out[13]: 12
```

```
In [14]: def add(x,y,z):
          c=x+y+z
          return c
          add(3,4)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[14], line 4
      2     c=x+y+z
      3     return c
----> 4 add(3,4)

TypeError: add() missing 1 required positional argument: 'z'
```

```
In [16]: def add(x,y):
          c=x+y
          return c
          print(add(3,4))
          def greet():
              return 'hello good morning'
          print(greet())
```

```
7
hello good morning
```

```
In [17]: def add(x,y):
          c=x+y
          return c
          def greet():
              return 'good morning'
          print(add(3,4))
          print(greet())
```

```
7
good morning
```

```
In [18]: def add(x,y):
          c=x+y
          return c
          def greet():
              return 'good morning'
          r1=add(3,2)
          r2=greet()
          print(r1)
          print(r2)
          print(type(r1))
          print(type(r2))
```

```
5
good morning
<class 'int'>
<class 'str'>
```

```
In [21]: def add(x, y):
          c = x + y
          print(c)

          def greet():
              print('hello good morning')

          r1 = add(3,4)
          r2 = greet()

          print(r1)
          print(r2)

          print(type(r1))
          print(type(r2))
```

```
7
hello good morning
None
None
<class 'NoneType'>
<class 'NoneType'>
```

```
In [22]: def greet():
          print('hello')
          print('good noon')
          def add(x,y):
              c=x+y
              print(c)
          def sub(x,y,z):
              d=x-y-z
              print(d)
          greet()
          add(5,4)
          sub(10,2,4)
```

```
hello
good noon
9
4
```

```
In [24]: def greet():
          print('hello')
          def add_sub(x,y):
```

```
c=x+y
d=x-y
print(c)
print(d)
greet()
add_sub(5,4)
```

```
In [25]: def greet():
          print('hello')

def add_sub(x,y):
    c=x+y
    d=x-y
    print(c)
    print(d)

r1 = greet()
r2 = add_sub(5,4)

print(type(r1))
print(type(r2))
```

```
hello
9
1
<class 'NoneType'>
<class 'NoneType'>
```

```
In [26]: def greet():
          return 'hello'
def add_sub(x,y):
    c=x+y
    d=x-y
    return c,d
r1=greet()
r2=add_sub(5,4)
print(r1)
print(r2)
print(type(r1))
print(type(r2))
```

```
hello
(9, 1)
<class 'str'>
<class 'tuple'>
```

```
In [27]: def greet():
          return 'hello'

def add_sub(x,y):
    c=x+y
    d=x-y
    return c
    return d

r1 = greet()
r2 = add_sub(5,4)

print(r1)
print(r2)
```

```
print(type(r1))
print(type(r2))
```

```
hello
9
<class 'str'>
<class 'int'>
```

```
In [36]: def greet():
         return 'hello'

         def add_sub(x,y):
             c=x+y
             d=x-y
             return c,d

         r1 = greet()
         r2,r3 = add_sub(5,4)

         print(r1)
         print(r2)
         print(r3)

         print(type(r1))
         print(type(r2))
         print(type(r3))
```

```
hello
9
1
<class 'str'>
<class 'int'>
<class 'int'>
```

```
In [39]: def add(a,b):
         c=a+b
         print(c)
         add(5,6)
```

```
11
```

```
In [ ]: 1-positional argument
```

```
In [40]: def person(name,age):
         print(name)
         print(age)
         person('nit')
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[40], line 4
      2     print(name)
      3     print(age)
----> 4     person( )

TypeError: person() missing 1 required positional argument: 'age'
```

```
In [42]: def person(name):  
        print(name)  
        print(age)  
        person('nit',23)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[42], line 4  
      2     print(name)  
      3     print(age)  
----> 4 person(    ,23)  
  
TypeError: person() takes 1 positional argument but 2 were given
```

```
In [43]: def person(age, name):  
        print(name)  
        print(age)  
  
        person('nit', 23)
```

```
23  
nit
```

```
In [44]: def person(age,name):  
        print(name)  
        print(age+1)  
        person('nit',23)
```

```
23  
  
-----  
TypeError                                Traceback (most recent call last)  
Cell In[44], line 4  
      2     print(name)  
      3     print(age+1)  
----> 4 person(    ,23)  
  
Cell In[44], line 3, in person(age, name)  
      1 def person(age,name):  
      2     print(name)  
----> 3     print(age+1)  
  
TypeError: can only concatenate str (not "int") to str
```

```
In [ ]: 2-keyword argument
```

```
In [46]: def person(age,name):  
        print(name)  
        print(age+1)  
        person('nit',23)
```

```
23
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[46], line 4
      2     print(name)
      3     print(age+1)
----> 4     person(      ,23)

Cell In[46], line 3, in person(age, name)
      1 def person(age,name):
      2     print(name)
----> 3     print(age+1)

TypeError: can only concatenate str (not "int") to str

```

```

In [47]: def person(age,name):
          print(name)
          print(age+1)
          person(name='honey',age=23)

```

```

honey
24

```

```

In [48]: def person(age, name, mob):
          print(name)
          print(age+1)
          print(mob)

          person(name = 'nit', age = 23, 12345)

```

```

Cell In[48], line 6
      person(name = 'nit', age = 23, 12345)
                                     ^
SyntaxError: positional argument follows keyword argument

```

```

In [49]: def person(age, name, mob):
          print(name)
          print(age+1)
          print(mob)

          person(name = 'nit', age = 23, mob = 12345)

```

```

nit
24
12345

```

```

In [51]: def person(age, name, mob):
          print(name)
          print(age)
          print(mob)

          person('nit',23,12345)

```

```

23
nit
12345

```